Affine constructions: complete manuscript costs

Human Compression project

The source is Hou, Feng and Leemans, *Regular 3-polytopes of order* $2^n p$, arXiv:2001.02945v2, dated 14 July 2024. We compare its complete seventeen pages with our complete seven-page common paper and nine-page application. The exact admitted principal contract has thirty-three rows; this measurement changes none of its mathematical conclusions.

The combined written comparison. Using calibrated NFC characters without whitespace, the costs are

Complete document	Characters
Source L012	32 767
Common theory U_{Affine}	19 070
Application AF1	22 638
Whole program appendix	17 250

The combined written ratio is

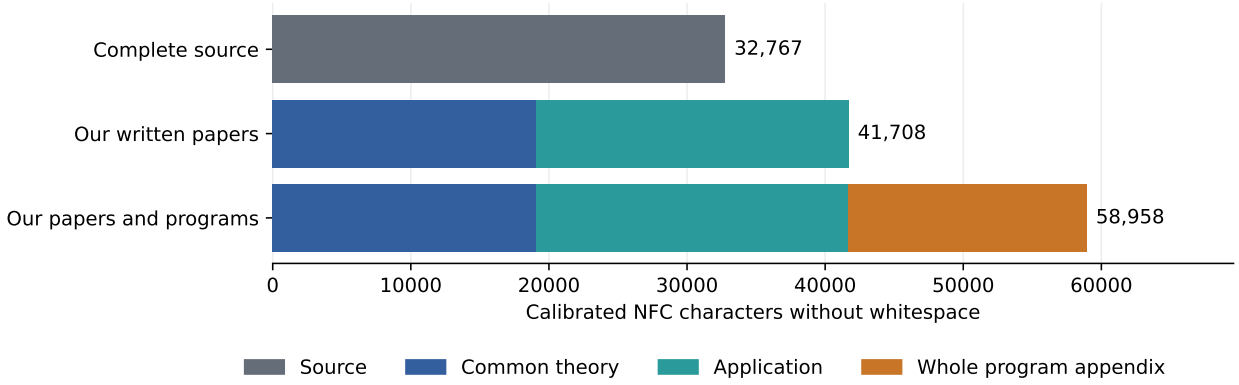
$$R_{\text{written}} = \frac{32\,767}{19\,070 + 22\,638} = \frac{32\,767}{41\,708} = 0.785629.$$

Thus the complete written replacement is longer than the source by this measure. The common paper is charged in full in this topic, including arguments also available in another topic. It is not discounted because they were previously proved or counted elsewhere.

The result after acquiring the common material. Once the whole common paper and all six complete required programs are available, the source/application ratio is

$$R_{\text{conditional}} = \frac{32\,767}{22\,638} = 1.447434.$$

This is a shorter local application. It does not make the common paper free in the combined comparison. Titles, context, references, complete proofs and all local hypotheses remain in every measured document.



The complete program boundary. The eight-page appendix prints six whole mathematical programs, including their embedded diagnostics. It is required material in the accepted scientific edition. Adding it once gives

$$R_{\text{programs}} = \frac{32\,767}{41\,708 + 17\,250} = \frac{32\,767}{58\,958} = 0.555769.$$

The six complete raw files cost 16 682 characters. Substituting this representation for the printed appendix gives

$$R_{\text{raw}} = \frac{32\,767}{58\,390} = 0.561175.$$

The two program representations are alternatives. They are never both added to the denominator, and no diagnostic fragment is pruned.

Three auxiliary scripts are inventoried separately: the replay driver, the checks for changed input files and the finite hypothesis diagnostics. They cost 16 581 characters in the same NFC coordinate. The original scientific metadata lists two different pairs; the inventory records their union of all three files without changing the scientific edition. The three supporting Lean files cost 7 958 characters and are separately recorded as formal validation. These costs are not silently added to a mathematical manuscript or to the program appendix.

What is included and what remains unknown. The numerator is the whole source manuscript for the admitted principal contract. It does not assert recovery of every auxiliary assertion. General Problem 1.1, the separate full Table 1 census and the ancillary uniqueness claim for the two order-384 seeds remain outside that contract. The source sign correction and all retained external theorems keep their existing attribution and qualifications. This measurement adds no credit for another source or a solved question.

The independent input review counted all forty-one physical PDF pages. Only page numbers and repeated running headers are removed. It checked 114 glyph substitutions against the actual encoded bytes and embedded fonts, and inspected every proposed furniture exclusion. Printed hyphens, formula ordering and extraction spacing are preserved. The same complete costs are supplied for three extraction policies, NFC and NFKC, and five length units: thirty coordinates in all. The CSV tables contain ninety combined fractions and thirty conditional fractions.

The source's 163 inline graphic marks have an unknown graphic and layout cost. Its original program files are unavailable. Expanded external proofs, the SmallGroups catalogue and the construction of exact group algorithms also have unknown acquisition costs. None is assigned zero. Consequently these are declared text and program comparisons, not a measure of all visible mathematical information or research effort.

Nine existing Lean declarations verify specified quotient, period and sign statements. They do not formalize the entire source contract. Paired proof dependencies and exclusive passage costs are still pending. The numerical packet awaits independent arithmetic and reader review; its input boundary has already been independently accepted.